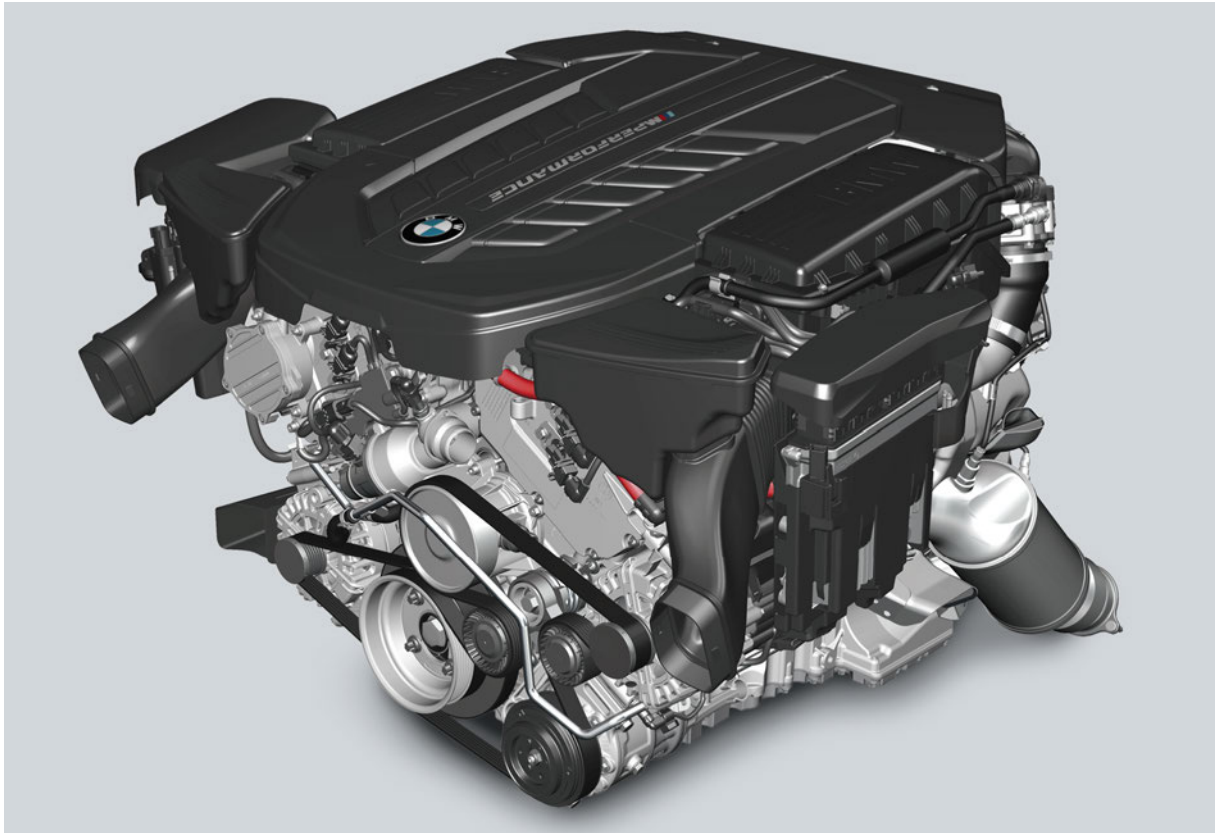


**Technical training.**  
**Product information.**

## **N74TU2 Engine**



**BMW Service**

Edited for the U.S. market by:  
**BMW Group University**  
**Technical Training**

ST1925

4/1/2019

# General information

## Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



---

Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

---

## Information status: November 2018

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

The information contained in the training course materials is solely intended for participants in this training course conducted by BMW Group Technical Training Centers, or BMW Group Contract Training Facilities.

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For changes/additions to the technical data, repair procedures, please refer to the current information issued by BMW of North America, LLC, Technical Service Department.

This information is available by accessing TIS at [www.bmwcenternet.com](http://www.bmwcenternet.com).

## Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application
- Aftersales Information Research (AIR)

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# N74TU2 Engine

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# N74TU2 Engine

## 1. Introduction

With the BMW M760i xDrive the revised 12-cylinder N74TU2 engine with M-Performance TwinPower turbo technology will be brought to market in March of 2019 in the G12 LCI.

The N74TU2 engine is the successor to the N74TU engine, but has many technological similarities. This results in an increase in the fuel injection pressure from 200 bar to 350 bar, making use of a new Digital Motor Electronics system (DME). Furthermore, the N74TU2 engine contains additional changes to details in the engine electrical system.

Only the differences between the N74TU2 engine and N74TU engine are described in this document.

### 1.1. Models

Models with a N74TU2 engine for launch in March 2019:

| Model            | Series  | Engine   | Power in kW/HP | Torque in Nm/lb-ft/rpm |
|------------------|---------|----------|----------------|------------------------|
| BMW M760i xDrive | G12 LCI | N74B66U2 | 448/600        | 850/627/1550-5000      |

### 1.2. History

BMW 12-cylinder gasoline engines:

| Engine   | Model        | Series | Displacement in cc | Power in kW/hp | Torque in Nm | Used as of |
|----------|--------------|--------|--------------------|----------------|--------------|------------|
| M70B50   | 750i         | E32    | 4988               | 220/300        | 450          | 05/87      |
| M70B50   | 850i         | E31    | 4988               | 220/300        | 450          | 04/90      |
| M70B50   | 750i         | E32    | 4988               | 220/300        | 450          | 09/90      |
| M73B54   | 750i         | E38    | 5379               | 240/326        | 490          | 09/94      |
| M73B54   | 850Ci        | E31    | 5379               | 240/326        | 490          | 09/94      |
| N73B60   | 760i         | E65    | 5972               | 327/445        | 600 rpm      | 09/02      |
| N73B60   | 760Li        | E66    | 5972               | 327/445        | 600 rpm      | 09/02      |
| N74B60   | 760Li        | F01    | 5972               | 400/544        | 750          | 09/09      |
| N74B60   | 760Li        | F02    | 5972               | 400/544        | 750          | 09/09      |
| N74B66U1 | M760i xDrive | G12    | 6592               | 448/600        | 800          | 11/16      |

# N74TU2 Engine

## 1. Introduction

### 1.3. Technical data

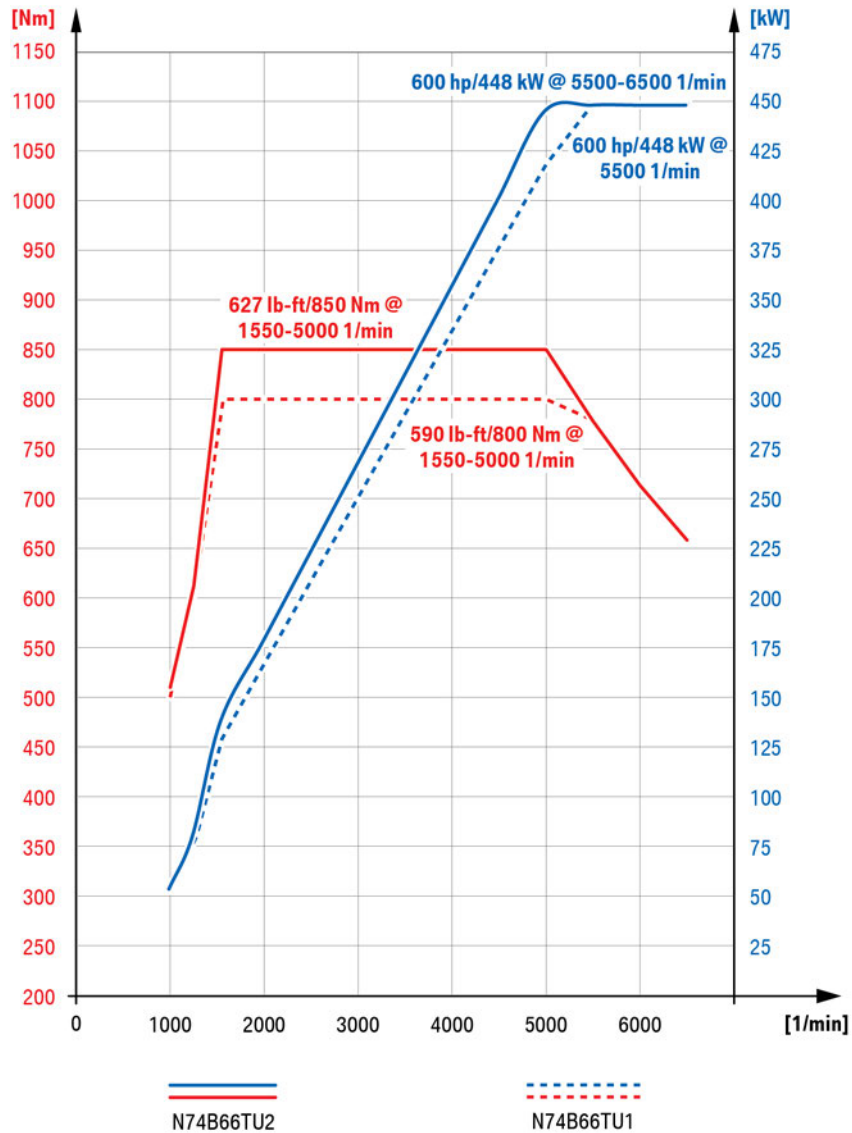
|                                                              |                     | <b>G12/M760i xDrive<br/>N74B66U1</b>    | <b>G12/M760i xDrive<br/>N74B66U2</b>    |
|--------------------------------------------------------------|---------------------|-----------------------------------------|-----------------------------------------|
| Design                                                       |                     | V12 60°                                 | V12 60°                                 |
| Firing order                                                 |                     | 1-7-5-11-3-9-<br>6-12-2-8-4-10          | 1-7-5-11-3-9-<br>6-12-2-8-4-10          |
| Displacement                                                 | [cc]                | 6592                                    | 6592                                    |
| Bore hole/Stroke                                             | [mm]                | 89/88.3                                 | 89/88.3                                 |
| Power output<br>at engine speed                              | [kW/hp]<br>[rpm]    | 448/600<br>5500                         | 448/600<br>5500-6500                    |
| Torque<br>at engine speed                                    | [Nm/lb-ft]<br>[rpm] | 800/590<br>1550-5000                    | 850/627<br>1550-5000                    |
| Cutoff speed                                                 | [rpm]               | 6500                                    | 6500                                    |
| Compression ratio                                            | [ε]                 | 10.0                                    | 10.0                                    |
| Distance between<br>cylinders                                | [mm]                | 98                                      | 98                                      |
| Valves per cylinder                                          |                     | 4                                       | 4                                       |
| Intake valve diameter                                        | [mm]                | 33.2                                    | 33.2                                    |
| Exhaust valve diameter                                       | [mm]                | 29                                      | 29                                      |
| Diameter of crankshaft<br>main bearing journals              | [mm]                | 65                                      | 65                                      |
| Diameter of crankshaft<br>connecting rod bearing<br>journals | [mm]                | 54                                      | 54                                      |
| Engine control                                               |                     | 2 x DME 8.C0                            | 2 x DME 8.CT                            |
| Acceleration 0-60 mph                                        | Seconds             | 3,6                                     | 3.6                                     |
| Maximum speed                                                | mph                 | 155                                     | 155                                     |
| Exhaust emission<br>standards                                |                     | Ultra Low Emission<br>Vehicle (ULEV) II | Ultra Low Emission<br>Vehicle (ULEV) II |

# N74TU2 Engine

## 1. Introduction

### 1.3.1. Full load diagram

The full load diagram shows the performance data of the N74TU2 engine.



N74TU2 engine, full load diagram in comparison to the N74TU1 engine

T018-0934\_2

# N74TU2 Engine

## 1. Introduction

### 1.3.2. Engine identification

#### Engine designation

In the technical documentation the engine designation is used to ensure clear identification of engines.

The N74TU2 engine market launch has the following specifications: N74B66U2.

The technical documentation also contains the short form of the engine designation N74TU2, which only indicates the engine type.

This is broken down as follows:

| Index | Explanation                                           |
|-------|-------------------------------------------------------|
| N     | BMW Group "New Generation"                            |
| 7     | 12-cylinder engine                                    |
| 4     | Engine with high-pressure injection and turbocharging |
| B     | Gasoline engine                                       |
| 66    | 6.6 liter capacity                                    |
| U     | Lower power stage                                     |
| 2     | Second revision                                       |

#### Engine identification and engine number

The engines have an identification mark on the crankcase to ensure unambiguous identification and classification. Engine identification is also necessary for approval by the authorities. The first 7 characters are relevant here. The N74TU2 engine identification corresponds to the new standard, whereby the first 6 characters match those of the engine designation. The seventh character is a sequential letter that can be used for different distinctions, such as the power stage or exhaust emission standard. A fixed correlation is not possible – though an "A" generally refers to the basic model.

The engine number is a sequential number that permits unique identification of individual engines. Engine identification and number are located on the crankcase behind the mounting bracket for the air conditioning compressor.



# N74TU2 Engine

## 1. Introduction

### 1.4. Overview

The following table lists the most important changes to the N74TU2 engine in comparison to the N74TU engine. More detailed information can be found in the thematic chapters.

| Assembly                    | N74TU engine                                                                                                                               | N74TU2 engine                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Cylinder heads              | Adaptation to the solenoid valve injectors                                                                                                 | Geometric adaptation of the cylinder heads (HDE 6)                                   |
| Camshafts                   |                                                                                                                                            | New intake camshafts with adapted cam shape (HDP 6)                                  |
| Cooling                     | Analog activation of the electric coolant pumps                                                                                            | LIN-Bus activation of the electric coolant pumps                                     |
| Exhaust gas after-treatment | Engine-proximate catalytic converters                                                                                                      | Engine-proximate catalytic converters                                                |
| Exhaust turbocharger        | Turbine housing as integrally-cast manifold / turbine housing unit.<br>Electromechanical wastegate valve actuator and air-diverter valves. |                                                                                      |
| Fuel preparation            | High-pressure injection with solenoid valve injectors HDEV 5.2                                                                             | High-pressure injection HDE 6 with up to 350 bar and solenoid valve injectors HDEV 6 |
| Engine control              | DME 8.C0                                                                                                                                   | DME 8.CT ( <b>not</b> cooled by coolant)                                             |

# N74TU2 Engine

## 2. Engine Mechanical

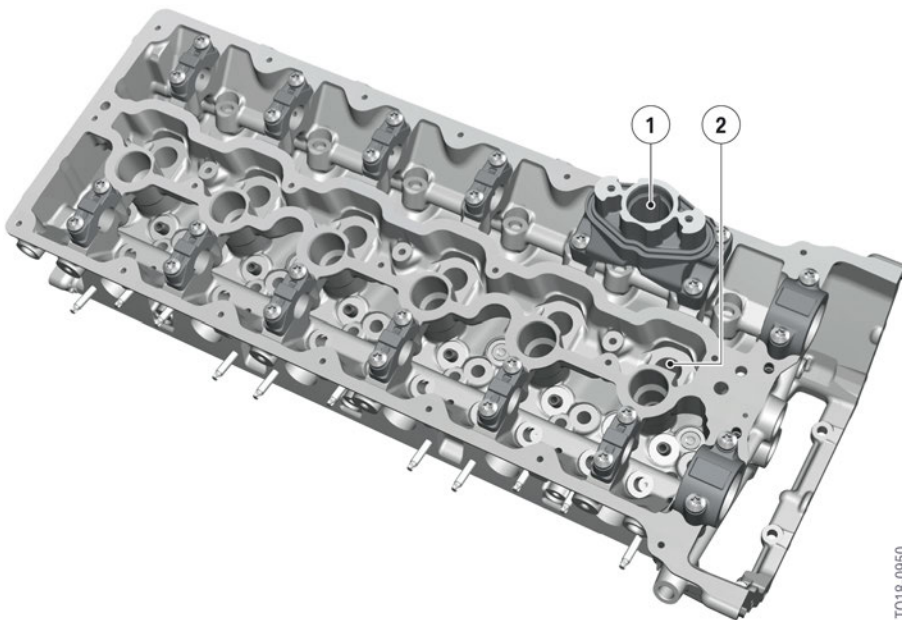
The engine mechanics of the N74TU2 engine conforms to a large extent to the engine mechanics of the N74TU engine. The engine mechanics of the N74TU2 include the following changes:

- Geometric adaptation of the cylinder head to the HDE 6 with up to 350 bar.
- Geometric adaptation of the intake camshaft to the HDE 6 with up to 350 bar.

### 2.1. Cylinder head

The design of the cylinder head in the N74TU2 engine is also characterized by the central location of the injector and spark plug in the combustion chamber. The cylinder head in the N74TU2 engine has the same design concept as that of the N74TU engine, but has been re-adapted for the N74TU2 engine.

To support the new HDEV 6 injectors, the holes in the cylinder head had to be adapted. Furthermore, the supports and roller tappets for the HDP 6 high pressure pump have been adapted to the cylinder head.



N74TU2 engine, cylinder head

| Index | Explanation                     |
|-------|---------------------------------|
| 1     | Mounting for high pressure pump |
| 2     | Injector hole                   |

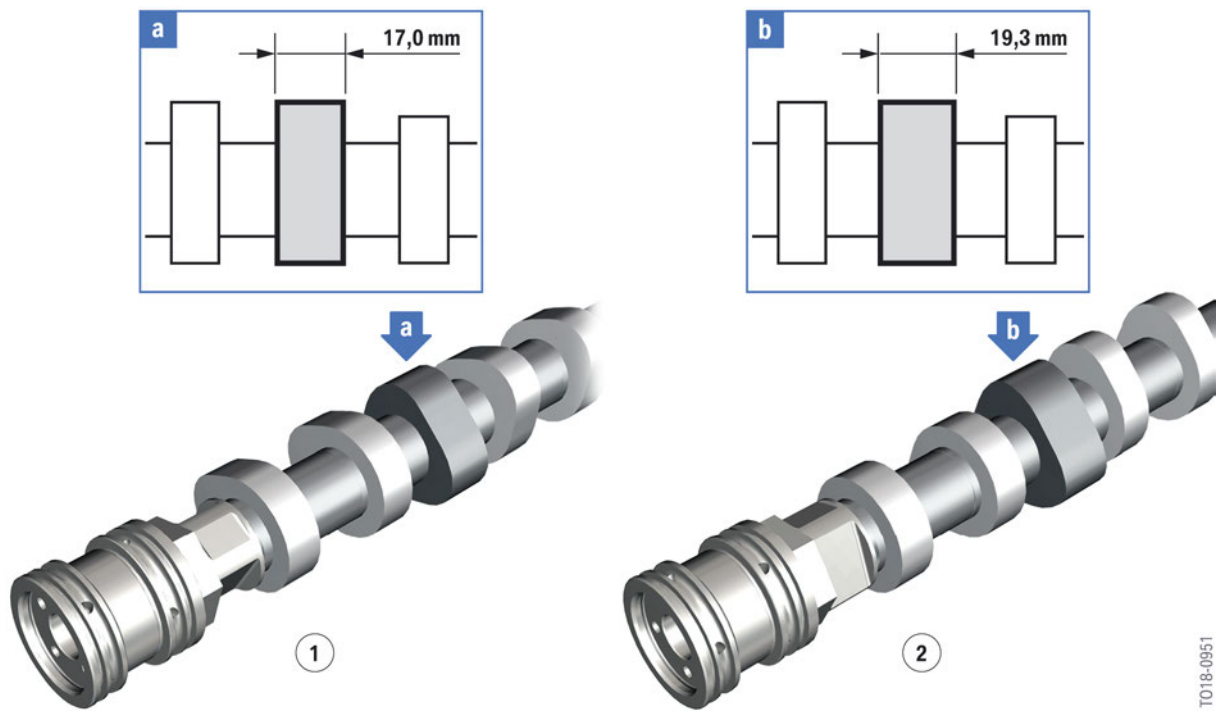
# N74TU2 Engine

## 2. Engine Mechanical

### 2.2. Camshafts

#### 2.2.1. Camshafts

The N74TU2 engine possesses the known camshafts in a lightweight construction. All cams are forced onto knurled points. The intake camshafts also each have one triple cam to drive the high pressure pumps. The geometry of the triple cam has been widened on the inlet side to comply with the requirements of the 350-bar high-pressure injection.



N74TU2 engine, intake camshaft comparison

| Index | Explanation                   |
|-------|-------------------------------|
| 1     | Intake camshaft N74TU engine  |
| 2     | Intake camshaft N74TU2 engine |

# **N74TU2 Engine**

## **3. Oil Supply**

The oil supply of the N74TU2 engine is the same as the oil supply of the N74TU engine.

# N74TU2 Engine

## 4. Cooling

Apart from some minor changes, the cooling system is the same as the cooling system of the N74TU engine. The following changes have been made to the cooling system in the N74TU2 engine:

- Connection of the electric coolant pumps to the Digital Motor Electronics (DME) via LIN bus. (N74TU engine = PWM)
- In the charge air cooling circuit, the cooling of the two control units of the Digital Motor Electronics (DME) has been discontinued.

# N74TU2 Engine

## 5. Fuel Preparation

For the N74TU2 engine, the high-pressure injection HDE 6 is used.

Changes were made in regard to the fuel injection pressure, which can now range up to 350 bar. Increasing the fuel injection pressure from 200 bar to 350 bar results in the following advantages:

- Improved fuel atomization
- More efficient mixture
- Reduced fuel wall film formation
- Shorter injection periods

These advantages due to the fuel injection pressure increase to 350 bar led to the following improvements:

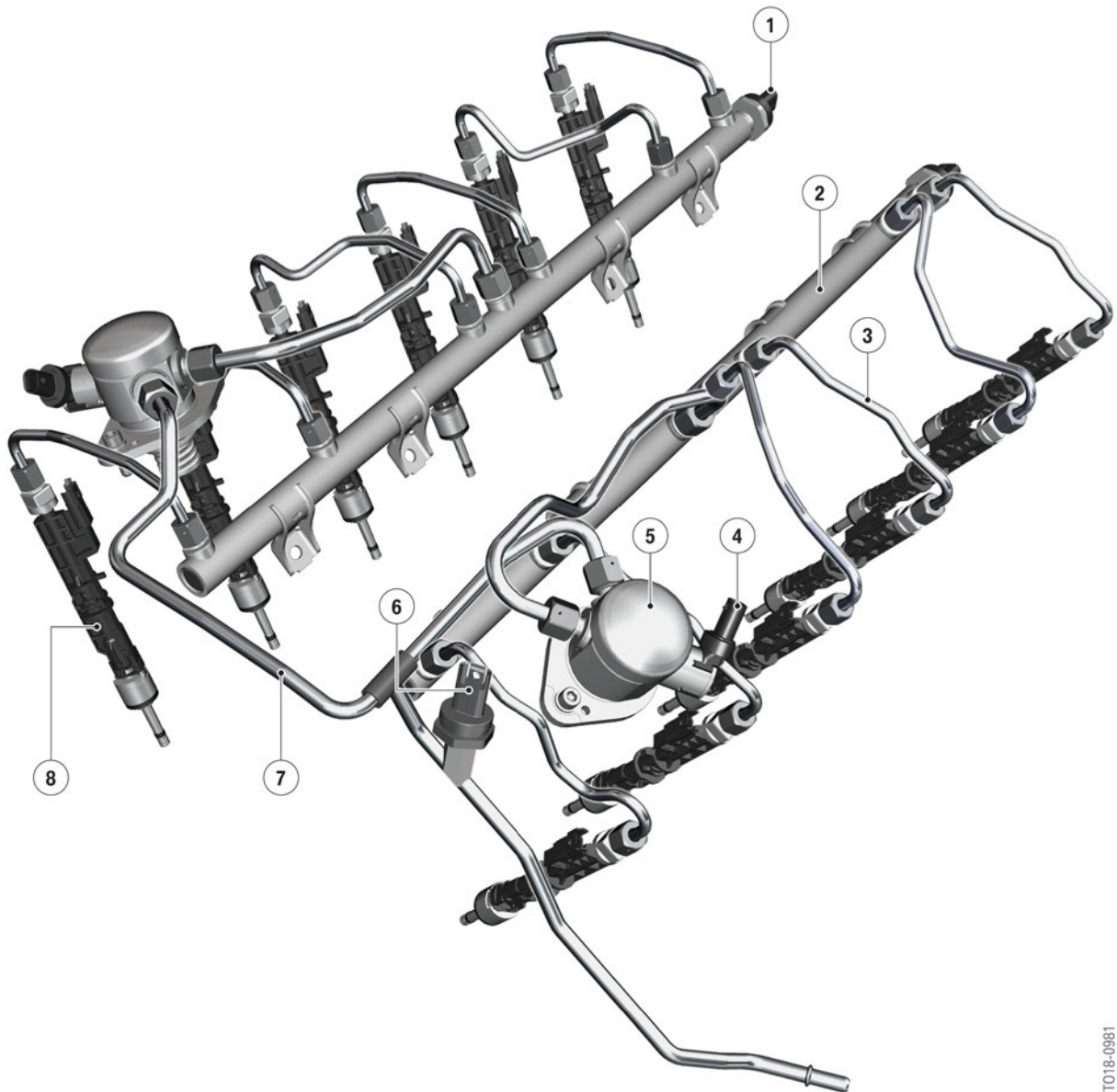
- Reduction in particle emissions
- Improved engine response under high engine load
- Improved throttle response with dynamic engine operation.

### 5.1. Overview

The following overview shows the fuel preparation of the N74TU2 engine. It essentially corresponds to the systems with direct fuel injection familiar in BMW models. Only the routing of the fuel feed line to the high pressure pumps has been modified and adapted.

# N74TU2 Engine

## 5. Fuel Preparation



N74TU2 engine, fuel preparation

TO18-0981

| Index | Explanation                         |
|-------|-------------------------------------|
| 1     | Rail pressure sensor                |
| 2     | Rail                                |
| 3     | High pressure line, rail - injector |
| 4     | Fuel quantity control valve         |
| 5     | High pressure pump                  |
| 6     | Fuel low-pressure sensor            |
| 7     | Fuel delivery line                  |
| 8     | Injector                            |

# N74TU2 Engine

## 5. Fuel Preparation

### 5.2. Fuel pump control

The electric fuel pump delivers the fuel from the fuel tank via the feed line to the high pressure pump at a primary pressure of 5.9 bar. The on-load speed control is effected via the Digital Motor Electronics.

### 5.3. High pressure pump

The well-known Bosch HDP 6 high pressure pump is used. This is a single-piston pump driven from the exhaust camshaft via a triple cam. So that sufficient fuel pressure is guaranteed in each load condition of the engine, a high pressure pump is used in the N74TU2 engine for each cylinder bank. The pump piston and the tappet have been modified on the high-pressure pump, along with material adaptations. Its structural shape is already known from the 4, 8 and 12-cylinder engines.

### 5.4. Injectors

Bosch high-pressure fuel injection valves with the designation HDEV 6 with CVO are used. Due to the fuel system pressure increase from 200 bar to 350 bar, the HDEV 6 solenoid valve injectors have been optimized using suitable materials and coatings.



# N74TU2 Engine

## 5. Fuel Preparation



T018-0872

N74TU2 engine, injector



Work on the fuel system is only permitted after the engine has cooled down. The coolant temperature must not exceed 40 °C / 104 °F. This stipulation must be observed without fail, as otherwise there is a risk of fuel being sprayed back on account of the residual pressure in the high-pressure fuel system.

When working on the high-pressure fuel system, it is essential to adhere to conditions of absolute cleanliness and to observe the work sequences described in the repair instructions. Even the slightest contamination and damage to the screw connections of the high-pressure lines can cause leaks.

When working on the fuel system of the N74TU2 engine, it is important to ensure that the ignition coils are not fouled with fuel. The resistance of the silicone material is greatly reduced by sustained contact with fuel. This may result in flashovers on the spark plug head and thus in misfires.

- Before making any modifications to the fuel system, without fail remove the ignition coils and protect the spark plug shaft against ingress of fuel by covering with a cloth.
- Prior to a new installation of the solenoid valve injectors, the ignition coils must be disassembled and the highest possible level of hygiene ensured.
- Ignition coils heavily fouled by fuel must be replaced.

# N74TU2 Engine

## 5. Fuel Preparation

- The CVO function comprises the "injector" and "Digital Motor Electronics" (DME) system components. These components therefore have to be identified with the vehicle identification number in the EPC in the event of a replacement.
  - Where injectors and DME support the CVO function, the injection quantity compensation during the replacement of one of the components is omitted.
  - An excessive rotational angle at the injector shank, and excessive tensile and compressive forces during removal and installation can lead to damage and therefore leaks in the fuel system.
  - The stems of the solenoid valve injectors can only withstand a certain tensile force and a certain torque. When removing and installing the injectors it is essential to follow the specific procedure set out in the repair instructions, as otherwise the injectors may be damaged.
  - Due to the design, dirt particles, grains of sand, etc. may enter the shafts of the injectors and spark plugs during operation, particularly in dusty environments with poor road surfaces. Before disassembly, the shafts must be blown out with a sharp jet of compressed air from all possible angles and positions using the longest available lance. Once the injector or spark plug has been removed, any particles should also be cleared from the edge of the bore hole.
  - For any service work required, the current information and specifications in the documents in the Integrated Service Technical Application (ISTA) must be observed in each case.
-

# N74TU2 Engine

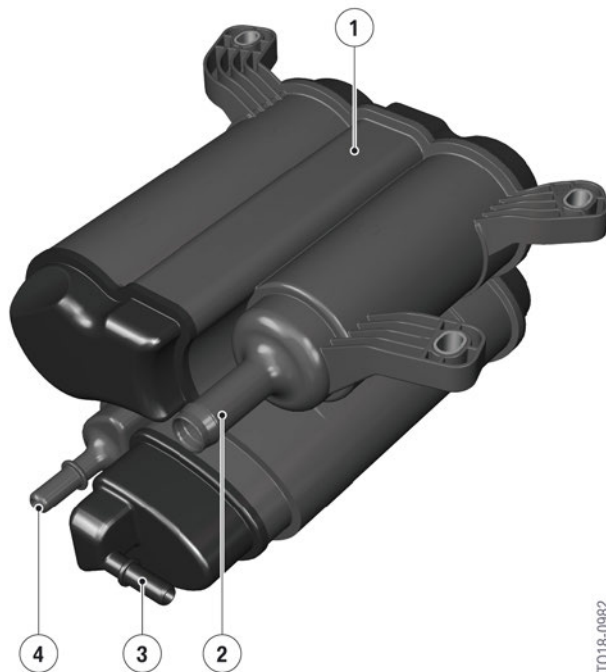
## 6. Fuel Supply

The fuel supply system corresponds to the fuel supply system of the N74TUengine, except for some minor changes. The following changes have been made to the fuel supply system in the N74TU2:

- Introduction of a larger active carbon filter to satisfy exhaust emission standards.

### 6.1. Carbon canister

The basic function of the carbon canister has not changed. The greater volume only serves to hold a greater quantity of fuel vapors and to achieve a longer intermediate storage before flushing by the tank vent valve.



N74TU2 engine; active carbon filter

T018-0982

| Index | Explanation               |
|-------|---------------------------|
| 1     | Carbon canister           |
| 2     | Ventilation to atmosphere |
| 3     | Tank ventilation          |
| 4     | To tank vent valve        |

# N74TU2 Engine

## 7. Engine Electrical System

### 7.1. Engine control unit

The Digital Motor Electronics (DME) is the computing and switching center of the engine control system. Sensors on the engine and on the vehicle deliver the input signals. The signals for activating the actuators are calculated from the input signals, the setpoint values calculated using a computing model in the DME control unit and the stored characteristic maps. The DME control unit activates the actuators directly or via relays.

The DME control unit is woken up via the wake-up line (terminal 15 Wake up).

The after-run begins once the terminal is switched to OFF. The adaptation values are stored during the after-run. The DME control unit uses a bus signal to signal its readiness to "go to sleep". When all the participating control units have signalled their readiness to "go to sleep", the bus master outputs a bus signal and the control units terminate communication 5 seconds later.

The printed circuit board in the DME control unit accommodates 2 sensors: a temperature sensor and an ambient pressure sensor. The temperature sensor is used to monitor the temperature of the components in the DME control unit. The ambient pressure is required for calculating the mixture composition.

#### 7.1.1. DME Generation 8

A new generation of Bosch engine control units is used in the N74TU2 engine. The 8th generation of engine electronics (DME) represents a joint control unit platform for petrol and diesel engines and is already used for modular engines. Its appearance is characterized by a uniform housing and a uniform connector strip. However, the hardware inside has been adapted to the various applications.

The control unit code (DME 8.CT.0) can be decoded as follows:

| Abbreviation Meaning |                                           |
|----------------------|-------------------------------------------|
| DME                  | Digital Motor Electronics                 |
| 8                    | Control unit generation                   |
| C                    | Number of cylinders as hexadecimal figure |
| T                    | Technical update                          |
| 0                    | Vehicle electrical system architecture    |

Number of cylinders as hexadecimal figure:

- 4 = 4-cylinder engine
- 6 = 6-cylinder engine
- 8 = 8-cylinder engine
- C = 12-cylinder engine.

Hardware variant:

- T = TU1 (technical revision 1)

# N74TU2 Engine

## 7. Engine Electrical System

Vehicle electrical system architecture:

- 0 = variant electrical system 2015
- 1 = variant electrical system 2018

Examples for petrol engines:

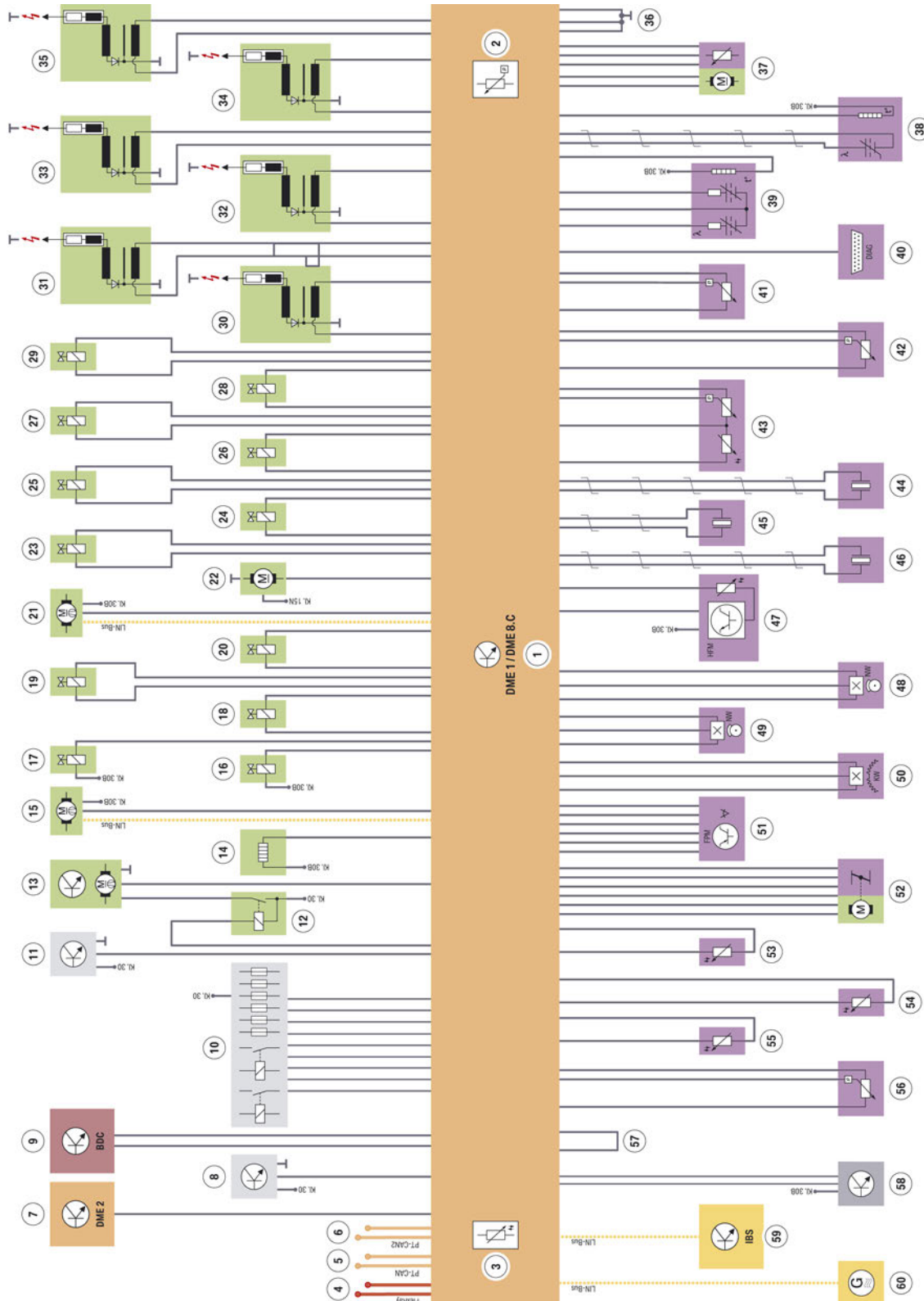
- DME 8.6.1 = B58 (variant electrical system 2)
- DME 8.8.0 = N63TU2 (variant electrical system 1)
- DME 8.8T.0 = N63TU3 (variant electrical system 1)
- DME 8.CT.0 = N74TU2 (vehicle electrical system variant 1).

This is why the N74TU2 engine features Bosch Digital Motor Electronics with the designation DME 8.CT.0. A separate engine control unit is fixed to the engine for every cylinder bank. The actuators and sensors of cylinder bank 1 are assigned to the DME 1 control unit; accordingly, the DME 2 control unit is responsible for the functions of cylinder bank 2. DME 1 is the primary control unit and receives all information concerning the entire engine, such as regarding the crankshaft sensor, and provides it to the DME 2 control unit directly or via the bus system. Due to the variety of sensors and actuators it was deemed necessary to use 2 control units.

# N74TU2 Engine

## 7. Engine Electrical System

### 7.1.2. Engine control, cylinder bank 1



N74TU2 engine, DME 1 system wiring diagram for DME 8.CT

# N74TU2 Engine

## 7. Engine Electrical System

| Index | Explanation                                                                    |
|-------|--------------------------------------------------------------------------------|
| 1     | Engine electronics DME 1 8.C                                                   |
| 2     | Ambient pressure sensor                                                        |
| 3     | Temperature sensor                                                             |
| 4     | FlexRay                                                                        |
| 5     | PT-CAN                                                                         |
| 6     | PT-CAN2                                                                        |
| 7     | Coupler CAN                                                                    |
| 8     | Tank leak diagnosis, Natural Vacuum Leak Detection (NVLD)                      |
| 9     | Body Domain Controller (BDC)                                                   |
| 10    | Power distribution box                                                         |
| 11    | Fuel pump control module (EKPS)                                                |
| 12    | Relay for electric fan                                                         |
| 13    | Electric fan                                                                   |
| 14    | Data-map thermostat                                                            |
| 15    | Electric coolant pump, charge air cooler                                       |
| 16    | Oil pressure control valve                                                     |
| 17    | Blow-off valve                                                                 |
| 18    | Tank vent valve                                                                |
| 19    | VANOS solenoid valve, intake camshaft                                          |
| 20    | VANOS solenoid valve, exhaust camshaft                                         |
| 21    | Electric coolant pump, exhaust turbocharger                                    |
| 22    | Electrical exhaust flap controller (EAKS)                                      |
| 23    | Quantity control valve                                                         |
| 24-29 | Injectors (cylinders 1-6)                                                      |
| 30-35 | Ignition coils (cylinders 1-6)                                                 |
| 36    | Earth                                                                          |
| 37    | Electric wastegate valve controller                                            |
| 38    | Oxygen sensor LSF 4.2                                                          |
| 39    | Oxygen sensor LSU ADV                                                          |
| 40    | Diagnostic connector                                                           |
| 41    | Charging pressure sensor after throttle valve                                  |
| 42    | Rail pressure sensor                                                           |
| 43    | Charge air temperature and charging pressure sensor upstream of throttle valve |
| 44    | Knock sensors 1-2                                                              |

# N74TU2 Engine

## 7. Engine Electrical System

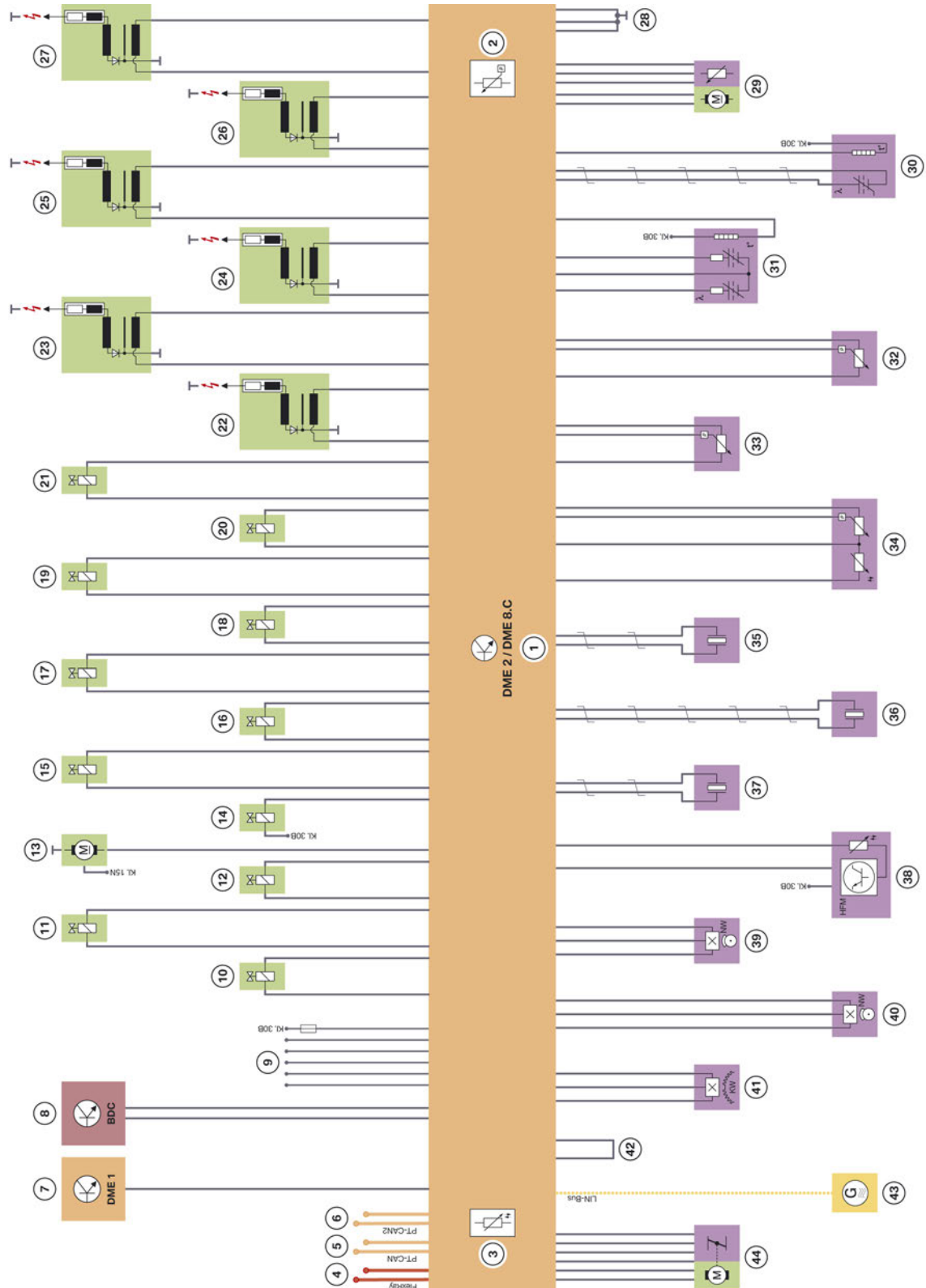
| Index | Explanation                                            |
|-------|--------------------------------------------------------|
| 45    | Knock sensors 3–4                                      |
| 46    | Knock sensors 5–6                                      |
| 47    | Hot film air mass meter 7                              |
| 48    | Camshaft sensor, intake camshaft                       |
| 49    | Camshaft sensor, exhaust camshaft                      |
| 50    | Crankshaft sensor (Y-cabling at DME 1 and DME 2)       |
| 51    | Accelerator pedal module                               |
| 52    | Throttle valve                                         |
| 53    | Engine temperature (sensor at housing of coolant pump) |
| 54    | Coolant temperature sensor at radiator outlet          |
| 55    | Oil temperature sensor                                 |
| 56    | Oil pressure sensor                                    |
| 57    | DME-1-/DME-2 coding                                    |
| 58    | Oil-level sensor                                       |
| 59    | Intelligent Battery Sensor (IBS)                       |
| 60    | Alternator                                             |



# N74TU2 Engine

## 7. Engine Electrical System

### 7.1.3. Engine control, cylinder bank 2



N74TU2 engine, DME 2 system wiring diagram for DME 8.CT

# N74TU2 Engine

## 7. Engine Electrical System

| Index | Explanation                                                                    |
|-------|--------------------------------------------------------------------------------|
| 1     | Engine electronics DME 2 8.C                                                   |
| 2     | Ambient pressure sensor                                                        |
| 3     | Temperature sensor                                                             |
| 4     | FlexRay                                                                        |
| 5     | PT-CAN                                                                         |
| 6     | PT-CAN2                                                                        |
| 7     | Coupler CAN                                                                    |
| 8     | Body Domain Controller (BDC)                                                   |
| 9     | Power distribution box                                                         |
| 10    | VANOS solenoid valve, intake camshaft                                          |
| 11    | VANOS solenoid valve, exhaust camshaft                                         |
| 12    | Quantity control valve                                                         |
| 13    | Electrical exhaust flap controller (EAKS)                                      |
| 14    | Blow-off valve                                                                 |
| 15    | Tank vent valve                                                                |
| 16-21 | Injectors (cylinders 7-12)                                                     |
| 22-27 | Ignition coils (cylinders 7-12)                                                |
| 28    | Earth                                                                          |
| 29    | Electric wastegate valve controller                                            |
| 30    | Oxygen sensor LSF 4.2                                                          |
| 31    | Oxygen sensor LSU ADV                                                          |
| 32    | Charging pressure sensor after throttle valve                                  |
| 33    | Rail pressure sensor                                                           |
| 34    | Charge air temperature and charging pressure sensor upstream of throttle valve |
| 35    | Knock sensors 7–8                                                              |
| 36    | Knock sensors 9–10                                                             |
| 37    | Knock sensors 11–12                                                            |
| 38    | Hot film air mass meter 7                                                      |
| 39    | Camshaft sensor, intake camshaft                                               |
| 40    | Camshaft sensor, exhaust camshaft                                              |
| 41    | Crankshaft sensor                                                              |
| 42    | DME-1-/DME-2 coding                                                            |
| 43    | Alternator                                                                     |
| 44    | Throttle valve                                                                 |

# N74TU2 Engine

## 7. Engine Electrical System



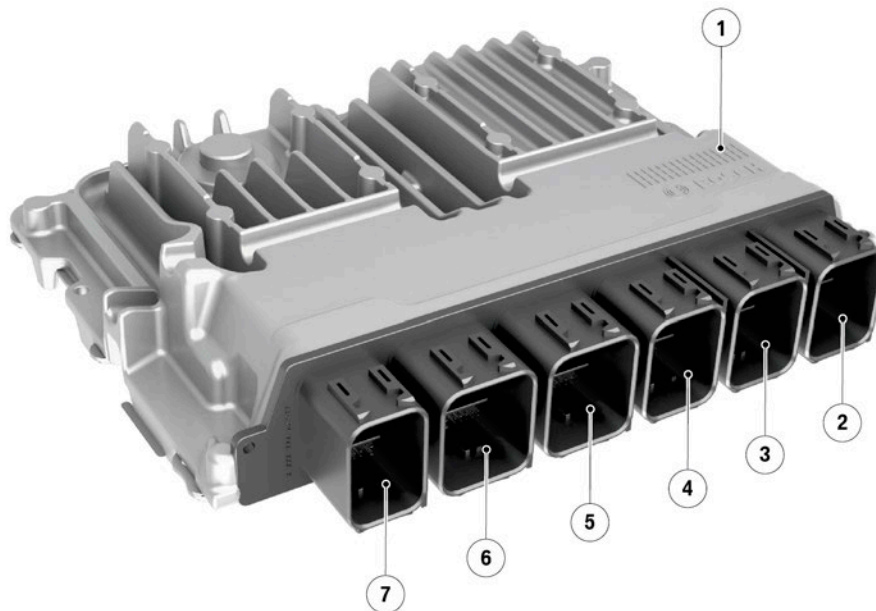
Do not attempt any trial replacement of control units.

Because of the electronic immobilizer, a trial replacement of control units from other vehicles must not be attempted under any circumstances.

The connector concept is identical to that on the modular engines and features a Nano MQS connector system (Micro Quadlock System). There is a logical division into 6 modules.



Measurements on the wiring harness may only be taken using measuring procedures approved by BMW. Use of the incorrect tools, such as measuring probes, can damage the plug-in contacts.



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N74TU2 engine, connections for DME 8.CT

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| 1     | Engine electronics DME 8.CT         |
| 2     | Module 600, injection and ignition  |
| 3     | Module 500, DME supply              |
| 4     | Module 400, Valvetronic servomotor  |
| 5     | Module 300, sensors and actuators 2 |
| 6     | Module 200, sensors and actuators 1 |
| 7     | Module 100, vehicle connection      |

